

Database Query Performance Profiler

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Use-Case Flow: Generate Weekly Digest

Preconditions

At least one week of slow query log data has been ingested and parsed, and one or more index recommendations exist in the system. The Scheduler is configured to trigger the digest job weekly.

Postconditions

A digest record is created and persisted with a unique digest ID and timestamp. The Backend Lead has been notified that the digest is available.

Main Success Scenario

1. The Scheduler triggers the weekly digest job at the configured time.
2. The system retrieves all flagged queries and index recommendations logged in the past seven days.
3. The system groups the recommendations by affected table and ranks them by estimated performance impact.
4. The system compiles the digest, including query counts, recommended indexes, and priority levels for each entry.
5. The system persists the digest record with a timestamp and a unique digest ID.
6. The system sends the digest through the Notification Service to the Backend Lead.
7. The Backend Lead receives the notification and opens the digest.
8. The use case ends successfully.

Alternate Flow — 3a. No Recommendations Found

- 3a1. The system detects zero flagged queries in the seven-day window.
- 3a2. The system generates a digest noting that no issues were detected instead of a recommendations list.
- 3a3. The digest is still persisted and delivered per steps 5–7 of the main scenario.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse Postgres/MySQL EXPLAIN plans, highlight sequential table scans on large tables, and recommend optimal index column definitions.	High	Pass: Missing composite index identified on filtered columns. Fail: Query plan with sequential scan marked optimized.	Core function of the profiler; without index recommendations the tool provides no actionable value.
FR-002	Functional	The system shall ingest slow query logs from Postgres and MySQL sources at configurable intervals and normalize them into a common internal schema before analysis.	High	Pass: A batch of 100 mixed-format log entries is fully ingested and normalized within 2 minutes. Fail: Any entry is dropped or left in native format after ingestion completes.	Ingestion is the entry point of the pipeline; unreliable or partial ingestion invalidates every downstream recommendation.
FR-003	Functional	The system shall generate a weekly performance optimization digest summarizing all flagged queries and index recommendations from the preceding seven days, grouped by affected table.	High	Pass: The digest generated each Monday at 00:00 UTC contains every recommendation logged in the prior week, grouped by table. Fail: The digest omits recommendations from the period or fails to generate on schedule.	A consolidated weekly view lets the Backend Lead act on database health without continuously monitoring raw logs.
FR-004	Functional	The system shall allow the Database Administrator to mark an index recommendation as Approved, Dismissed, or Applied, and persist that status for future digests.	Medium	Pass: The DBA changes a recommendation's status and the updated status is reflected in the next digest. Fail: The status change is not saved or not reflected downstream.	Without a way to record actioned recommendations, the tool would repeatedly re-flag issues the DBA has already resolved.
FR-005	Functional	The system shall notify the Backend Lead through the configured Notification Service when a new weekly digest becomes available.	Medium	Pass: A notification is delivered within 5 minutes of digest generation. Fail: The digest is generated but no notification is sent, or delivery exceeds 5 minutes.	A digest that is generated but not surfaced to its intended reader fails to change any behavior.
NFR-001	Non-functional	Slow query log ingestion must handle up to 5,000 log records per minute without dropping entries or slowing the host database.	High	Pass: Benchmarking tests confirm target latency and no measurable impact on host database performance under simulated peak load.	The profiler must never itself become a source of database performance degradation.
NFR-002	Non-functional	The system shall restrict recommendation approval, dismissal, and apply actions to authenticated users holding the Database Administrator role.	High	Pass: An attempted status change from a non-DBA account is rejected with an authorization error. Fail: A non-DBA account is able to change a recommendation's status.	Index changes affect production database performance, so only authorized personnel should be able to approve or apply them.

UML USE-CASE DIAGRAM

